

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
ACADEMIC YEAR 2021/2022
APRIL/MAY EXAMINATION
AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

WEDNESDAY, 18 MAY 2022

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE (DATA SCIENCE)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 1**

- a) The *Turing Test*, originally known as imitation game, was introduced by Alan Turing in 1950. Explain the purpose of *Turing Test* and how it works. (4 marks)
- b) Artificial intelligence (AI) is mainly concerned with rational action. An ideal intelligent agent performs actions to achieve its goals.
- (i) Explain what intelligent agent is in AI. (3 marks)
- (ii) You are required to design an automated taxi driving system, list the task environment and provide an example of each of the components. (4 + 4 marks)
- c) Problem formulation is an important phase when designing a problem-solving agent. List the **FIVE (5)** components of a formulated problem and briefly describe each of them. (10 marks)

[Total: 25 marks]

Question 2

- a) **Figure 1** shows some interconnected cities in Country X. The distance (in kilometer) from one city to another is shown on the arc. Assume that you are building a GPS system that examines all the possible ways and suggest the shortest path to reach the destination, L from A. Apply **Breath-First Search (BFS)** and **Depth-First Search (DFS)** to:

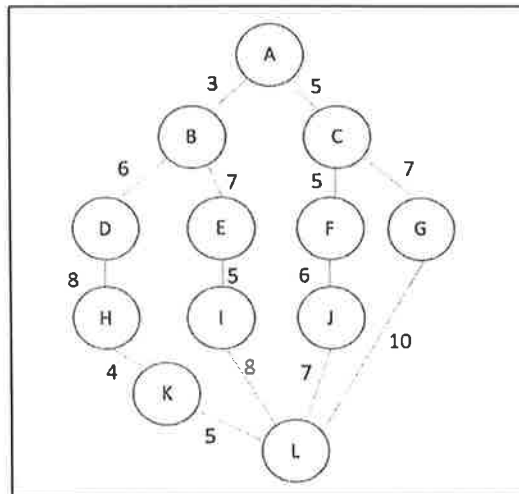


Figure 1: Interconnected cities in Country X

- (i) Identify the node expansion sequence from the initial state to the goal in alphabetical order. (1 + 1 marks)
- (ii) Identify the solution path. (1 + 1 marks)
- (iii) Identify the path cost of the solution. Show the working steps. (2 + 2 marks)

Question 2 a) (Continued)

- (iv) Evaluate the performances of DFS and BFS in terms of completeness and optimality based on the solution. (2 + 2 marks)
- b) A game can be formally defined as a kind of search problem with five elements.
- (i) Identify the **FIVE (5)** elements and briefly describe each of them. (5 marks)
- (ii) Compute the minimax value for the nodes (A to F) in the game tree shown in **Figure 2**. (6 marks)

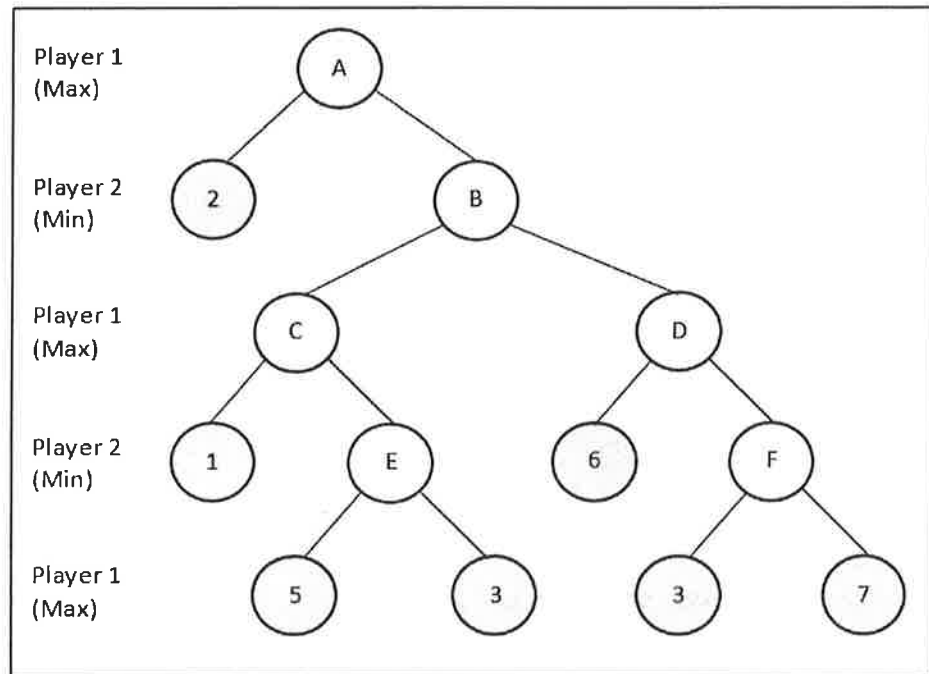


Figure 2: Game tree

- (iii) Identify Player 1's minimax decision. To answer this question, you are required to redraw the entire game tree and highlight the path/arcs. Justify the answer. (2 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3**

- a) Assuming you oversee schedule for computer science classes in College X. There are a total of 5 classes in a week. Two lecturers will be teaching the classes and each lecturer can only teach one class at a time.

The classes are:

Class 1: Object-oriented Programming: 9:00-10:00am

Class 2: Software Engineering: 9:30-10:30am

Class 3: Artificial Intelligence: 10:00-11:00am

Class 4: Machine Learning: 10:00-11:00am

Class 5: Computer Vision: 10:30-11:00am

The lecturers are:

Lecturer A, available to teach Class 3, 4.

Lecturer B, available to teach Class 1, 2, 3, and 5.

Formulate this problem as a Constraint Satisfaction Problems (CSP) and identify:

- (i) The variables (one per class) and the domains for each of the variable. (5 marks)
 - (ii) The constraints. (6 marks)
 - (iii) Construct the constraint graph according to your CSP. (8 marks)
- b) Explain what *Backtracking* search is and why it is called *Backtracking*. (4 + 2 marks)
- [Total: 25 marks]

Question 4

- a) Humans know things; and what we know helps us do things. Knowledge-based agents benefit from knowledge expressed in general form, combining information suits various purposes. With the aid of a diagram, explain what knowledge-based agents are. (3 + 3 marks)
- b) The syntax of *Propositional Logic* defines the allowable sentences for the knowledge representation. There are two types of propositions, which are *Atomic Propositions* and *Complex Propositions*. Differentiate between *Atomic Propositions* and *Complex Propositions*. Provide an example to illustrate your understanding. (4 marks)
- c) There are five logical connectives in *Proposition Logic*. Let:
- E = Slimer is eating
 - H = Slimer is hungry
 - HA = Slimer is happy

Translate the sentences below into proposition formulas with the correct logical connectives.

- (i) Slimer is eating and not hungry. (1 mark)
- (ii) If Slimer is eating, then Slimer is not hungry. (1 mark)

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4 c) (Continued)**

- (iii) Slimer is hungry but happy. (1 mark)
- (iv) It is false that Slimer is hungry and eating but not happy. (1 mark)
- d) Identify if $(p \Rightarrow q) \vee (p \Rightarrow \neg q)$ is valid using the truth tables method. (7 marks)
- e) Entailment can be done by theorem proving. Validity and satisfiability are among the concepts related to entailment. Describe validity and satisfiability with the aid of an example for each. (4 marks)

[Total: 25 marks]